[AMPLITUDE CHECKLIST]

In Unit 2, you learned that amplitude is defined as the distance the drum moves into the mat and that the impact force created by drum movement is the primary factor for creating density in an asphalt layer. You also learned that too much amplitude can damage an asphalt layer. Selecting the correct amplitude is a vital step during the planning process for any asphalt compaction application.

The compactor operator, quality control technician or supervisor need to create a checklist of the factors covered in Unit 3. Whether this is a mental checklist or a printed form, each factor must be considered. Following is a sample checklist.

AMPLITUDE CHECKLIST

Factor	Low Force	High Force
Layer Thickness	< 50 mm (2")	>50 mm (2")
Base Support	Rigid	Flexible
Oil Viscosity	Low	High
Aggregate Shape	Rounded	Angular
Ambient Temperature	High	Low

This checklist helps tie together all the compaction factors that have been covered in Unit Three and will help determine how much compaction force is needed for a particular application. Down the left side are the application factors that would tend to result in selecting medium-amplitude settings. Down the right side are the application factors that would tend to lead to the selection of medium- or high-amplitude settings.

As a rule, layer thickness less than 50 mm (2") would not accept high compaction energy. Therefore, amplitude settings in the range 0.25 mm to 0.6 mm (0.01 - 0.025") would typically be selected. When the asphalt layer is thicker than 50 mm (2"), the mat can accept more compaction energy and you would tend to select amplitude that is 0.6 mm (0.025") or higher.

The type of base being paved over has an effect on amplitude selection. If the base is rigid, like a milled surface or an existing asphalt surface, too much compaction energy will cause drum bouncing more easily. If you are paving over granular material or stabilized subgrade, that type of base is more yielding. You would tend to select higher amplitudes for that application because the flexible base will absorb some of the compaction energy.

You also need to consider the type of asphalt cement used in a particular mix design. If the asphalt cement has been modified with fibers or latex rubber, for example, the oil will have high viscosity, the mix will be considered stiff and will need higher forces. Unmodified asphalt cement has a lower viscosity and that mix will need lower compaction energy to achieve the required density.

Always consider the shape of the aggregate in the mix design. Most high-traffic, highway-class asphalt structures use mixes with all fractured faces.

The angular aggregates produce high internal friction. You would tend to select higher amplitude settings in order to move the rock and remove air voids.

Most street and parking lot mixes have aggregates with some rounded faces. Since the aggregate moves more easily, these mixes require lower compaction energy.

Climate conditions are the final factor in the checklist. On hot, sunny days, asphalt layers tend to retain their heat longer and since you have more time you can use lower forces.

When the ambient temperature is low and there are high winds, the asphalt layer will cool off rapidly. In order to reach density before the mat loses too much temperature, you would select the highest amplitude setting to try to hit density quickly. Following are two examples of how using the checklist can be beneficial.

[AMPLITUDE SELECTION EXERCISE I]

AMPLITUDE CHECKLIST

Factor	Low Force	High Force
Layer Thickness	40 mm (1.6")	–
Base Support	Milled Surface	–
Oil Viscosity	–	High
Aggregate Shape	–	Crushed
Ambient Temperature	–	10° C (50° F)

On this project, the paver is laying down a 40 mm (1.6") thick layer of dense-graded asphalt with polymer-modified asphalt cement, largest aggregate size 12.5 mm (1/2") and aggregate with all fractured faces. This is a single layer of asphalt being laid on a milled surface. Looking at the amplitude checklist, you'll see that two of the factors are on the lower force side, while three of the factors indicate higher force will be needed.

Does this mean that you should select a medium to high amplitude?

In this application, you would probably select a low to medium amplitude. The layer thickness, less than 50 mm (2"), and the rigid base would not accept high amplitude energy. Sometimes certain factors, especially layer thickness, are more important than other factors in the amplitude checklist. Selecting the correct amplitude is not always easy. Remember the general rule when selecting amplitude: Select the highest amplitude that the mat can accept without creating drum bouncing or damage to the mat.

[AMPLITUDE SELECTION EXERCISE 2]

AMPLITUDE CHECKLIST

Factor	Low Force	High Force
Layer Thickness	–	75 mm (3")
Base Support	–	Granular
Oil Viscosity	–	High
Aggregate Shape	–	Crushed
Ambient Temperature	32° C (90° F)	–

For this project, the amplitude choice is relatively easy. You are laying down 75 mm (3") of dense graded asphalt as the first lift on granular material. The mix design uses high-viscosity asphalt cement and 19 mm (3/4") crushed aggregate with fractured faces. It is a hot day. Four factors line up on the high force side. Only the high temperature shows on the low-force side. This asphalt layer will accept high amplitude for several reasons. First, the layer is fairly thick. Second, and very importantly, the ratio of layer thickness to maximum aggregate size is 4:1.

Summary: *Knowing the factors that affect asphalt compaction is a vital skill for operators, quality control technicians and supervisors. Much of the information about mix design is contained in the job mix formula that should be available from the asphalt plant. Items such as layer thickness and type of base for each layer will be shown on cross section plans. Only climate conditions need to be analyzed for each shift.*

If you gather enough information and know how to interpret it, you and your team can make better decisions when setting up the compaction process. If you rely only on past performance or experience, you may be ignoring new valuable information.

What you have learned about the forces of compaction and the factors that affect compaction will help you set up the compaction train. That is the subject of Unit 4.


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Unit 4
METHODS AND SPECIFICATIONS

Properly setting up the compaction process is the key to your project's success. Following are the appropriate procedures, whether using method or end-result specifications.



Cat CD54 working in the initial phase of compaction.

Units 2 and 3 covered the forces of compaction and other factors that affect asphalt compaction. Unit 4 deals with setting up a compaction train that meets the project requirements for production, density and smoothness.

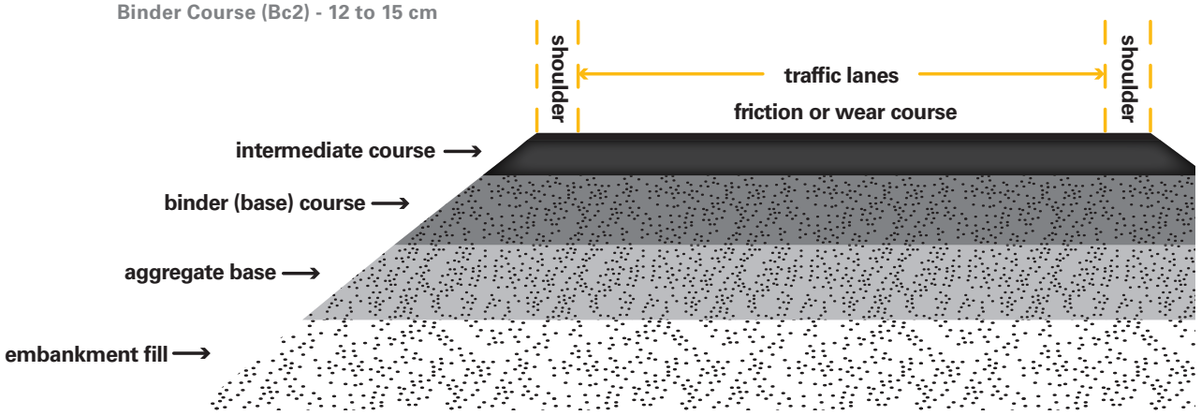
Unit 4 contains two distinct sections. The first section deals with setting up the compaction process with the guidance of a method specification that dictates type of equipment, working speed, number of passes, and other factors. The second section deals with setting up the compaction train under the assumption that an end result specification is in effect. In other words, the crew members and quality control personnel determine types of equipment and rolling patterns that will produce the desired density and match the production requirement, as well.

Section I: METHOD SPECIFICATIONS

The information in Section I is based on research done by the Central Laboratory for Bridges and Roads in Paris, France. Based on this widely-accepted research, the type of compaction equipment, number of passes, and working speed are specified for various types of bituminous pavement layers.

ROAD SECTION

Wearing Course thin lift (Wc1) - 3 to 5 cm
Wearing Course std lift (Wc2) - 6 to 9 cm
Intermediate Course (Ic) - 6 to 9 cm
Binder Course (Bc1) - 8 to 12 cm
Binder Course (Bc2) - 12 to 15 cm



[LAYER TYPE]

There are five categories of layers (courses) in this specification. A thin lift wear course (Wc1) has a thickness ranging from 3 to 5 cm. There is a standard lift wear course (Wc2) with a thickness ranging from 6 to 9 cm.

One binder (intermediate) course category is based on a thickness range from 6 to 9 cm.

There are two base course classifications. The standard base course (Bc1) is for thickness that ranges from 8 to 12 cm. The optional course (Bc2) ranges from 12 cm to 15 cm.

As explained in Unit 2, each layer type typically uses a different mix type and is laid to different thicknesses. Therefore, the amount of compaction force will be different for each layer.

[COVERAGE BASED ON DRUM WIDTH]

The relationship between drum width and the width of the layer is an important consideration. In order to match the production of the paving process, the drum width must be such that a compactor can cover the width of the layer in no more than three overlapping passes. The overlap is assumed to be a minimum of 15 cm. If the available compactor cannot cover the width of the layer in three overlapping passes, then more compactors must be added.

This chart is used as a guide to help determine how many rollers of a certain drum width would be required to cover various paving widths. The shaded areas indicate the need for at least one more roller, or a roller with wider drums, in order to match the production requirement. For example, one compactor with drums that are 170 cm wide will be effective on paving widths up to 4.5 m.

NUMBER OF ROLLERS REQUIRED BY ROAD WIDTH (COVERAGE)

Paving Width (meters)	Drum Width				
	150 mm	170 mm	200 mm	210 mm	300 mm
3.3	1	1	1	1	1
3.6	1	1	1	1	1
3.9	1	1	1	1	1
4.2	2	1	1	1	1
4.5	2	1	1	1	1
4.8	2	2	1	1	1
5.1	2	2	1	1	1
5.4	2	2	1	1	1
5.7	2	2	2	1	1
6	2	2	2	2	1
7	2	2	2	2	1
8	2	2	2	2	1
9	3	2	2	2	2

Note: *Compactors with drums offset to maximum width are recommended only when compacting thin wear courses.*

[COMPACTOR CLASSIFICATIONS]

In the French system, vibratory steel drum rollers are classified according to static linear load per drum and the amplitude of the drum. The classification formula is expressed as linear load in kilograms per centimeter multiplied by the square root of the rated amplitude.

Most vibratory compactors have more than one amplitude. Therefore, the classification can change when amplitude is changed on a compactor.

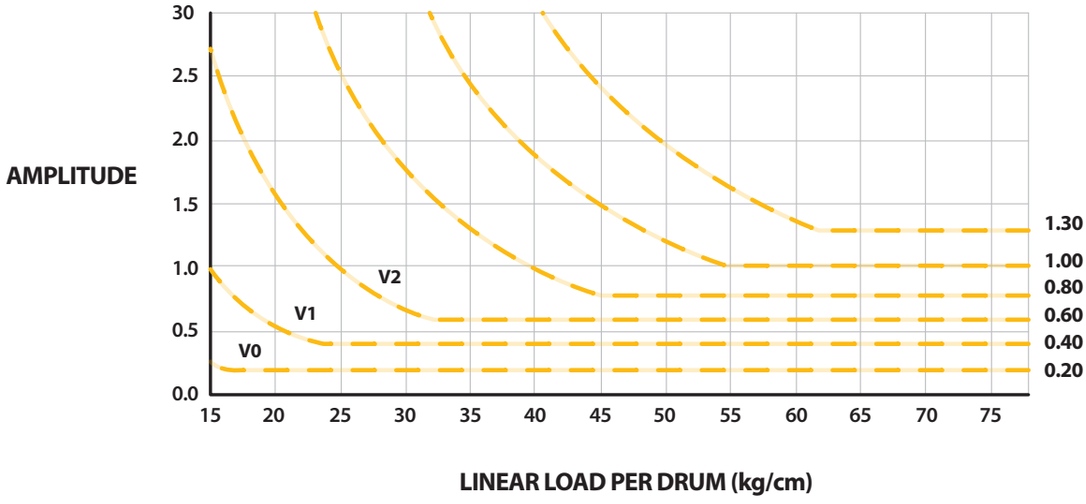
Vibratory Compactor Example

4680 kg (weight at the drum) / 170 cm (drum width) = 27.5 kg/cm
27.5 kg/cm x square root 0.62 mm = 21.7
21.7 = Class V1

[VIBRATORY COMPACTOR CLASSES]

The compactor in the chart below has three amplitude choices. Changing amplitude results in changing the classification of the compactor. The VO class will have a relatively low amount of compaction energy. The V1 class will have a mid-range amount of compaction energy. The V2 class will deliver a high amount of compaction energy.

Range	Low	Mid	High
amplitude (AO) (mm)	0.34	0.80	1.05
weight at drum (kg/cm)	27.50	27.50	27.50
weight x sq. root of AO =	16.04	24.60	28.18
classification	**VO**	**V1**	**V2**



This chart shows the ranges of the various compactor classifications based on linear load per drum and amplitude. For practical purposes, only three classification ranges are used in the method specification: V0, V1 and V2.

There are two classifications for pneumatic compactors. The formula for pneumatic compactor classification is written as total weight in tons divided by number of wheels.

Pneumatic Compactor Example

21 tons (machine weight) ÷ 7 (number of wheels) = 3 tons/wheel
3 tons / wheel = Class P1

Pneumatic Compactor Classes

Class	Weight Per Wheel
P0	Greater than 1.5 tons and less than 2.5 tons
P1	Greater than 2.5 tons and less then 4.0 tons

	Wc1		Wc2		Ic		Bc1		Bc2	
	passes	speed	passes	speed	passes	speed	passes	speed	passes	speed
class P0	18	6								
class P1	14	6	20	6						
class V2			4	4	5	4	15	4	25	3
class V1	4	4	8	4	7	4				
class V0	7	4								

Note: *Speeds shown in km/h.*

[SELECTING MACHINE CLASSIFICATIONS]

The aim of the method specification is to blend the information regarding machine classification and layer type into a reference chart that will guide the user to selection of machine type, number of passes and working speed. Across the top of the chart are the five types of layers: thin wear course (Wc1); thick wear course (Wc2); intermediate course (Ic); standard base course (Bc1); and thick base course (Bc2).

Machines classes are shown on the left side of the chart. Light pneumatic compactor is P0. Heavy pneumatic compactor is P1. The three vibratory compactor classes range from the highest vibratory energy (V2) through the mid-range vibratory energy (V1) and ending with the lowest vibratory energy (V0).

For thin wear courses (Wc1), the P0 class pneumatic compactor should make 18 passes at a speed of 6 kilometers per hour. If a P1 class pneumatic compactor is used, that machine should make 14 passes at 6 kilometers per hour. A class V1 vibratory compactor must complete four passes at 4 kilometers per hour. A class V0 vibratory compactor must complete seven passes at 4 kilometers per hour. Note that V2 classes are not permitted on thin wear courses.

On thick wear courses (Wc2), P0 class pneumatic compactors are not allowed. P1 pneumatic compactors will complete twenty passes at 6 kilometers per hour. V2 class vibratory compactors are directed to complete four passes at 4 kilometers per hour. V1 class vibratory compactors must complete eight passes at 4 kilometers per hour. The chart provides specific directions for each machine class on each type of pavement layer.

Other tools, such as the Cat® Interactive Production Calculator, can help determine the number of compactors that will be needed based on the paving speed, the width of the layer and the drum width of the compactor.

COMPACTION CALCULATOR

Trucking	Roller Model:	[*Click to Select Another Model*]	[**CB534D**]	
Paver Speed	Course Type:	[*Click to Select Course Type*]	[**Wearing Course Std Lift 0/10 - 6 to 9 cm**]	
Compaction	**General Inputs**			
Material Used	Production Rate of Hot Plant:		[**160**]	*tonnes/hr*
Slope	Paving Thickness:		[**65**]	*mm*
	Paving Width:		[**4.00**]	*meter*
Thickness	Actual Drum Width:		[**170**]	*cm*
Job Summary	[*Click to Change Roller Efficiency Rate*]		[**77**]	*%*
Legal	Actual Roller Speed:		[**4**]	*km/h*
	Total Number of Passes:		[**4**]	
	Number of Roller Needed (based on width):		[**1**]	
	Number of Roller Needed (based on speed):		[**1**]	
Exit	**Number of Roller Needed:**		[**1**]	

One version of the Cat Interactive Production Calculator has been programmed with the formulas and requirements of the Central Laboratory for Bridges and Roads in Paris, France method specification for compactor selection and rolling patterns. In the first example, the production rate is 160 tonnes per hour. The project has a standard thickness wear course being paved at a depth of 65 mm and a width of 4 meters. The number of passes and the working speed are selected automatically based on the charts from the method specification. The program has calculated that one compactor with a drum width of 170 cm will be able to match the requirement for coverage and the requirement for production. What would happen if the production rate were increased?



COMPACTION CALCULATOR

- Trucking
- Paver Speed
- **Compaction**
- Material Used
- Slope
- Thickness
- Job Summary
- Legal

Roller Model: [*Click to Select Another Model*] [**CB534D**]

Course Type: [*Click to Select Course Type*] [**Wearing Course Std Lift 0/10 - 6 to 9 cm**]

General Inputs

Production Rate of Hot Plant:	[**200**]	*tonnes/hr*
Paving Thickness:	[**65**]	*mm*
Paving Width:	[**4.00**]	*meter*
Actual Drum Width:	[**170**]	*cm*
[*Click to Change Roller Efficiency Rate*]	[**77**]	*%*
Actual Roller Speed:	[**4**]	*km/h*
Total Number of Passes:	[**4**]	
Number of Roller Needed (based on width):	[**1**]	
Number of Roller Needed (based on speed):	[**2**]	
Number of Roller Needed:	[**2**]	

- Exit

When the hourly production rate is increased from 160 tonnes per hour to 200 tonnes per hour, the calculation shows that two compactors will be needed to match the increased production. Remember, the calculator is programmed to select the working speed and the number of passes based on course type and thickness.



COMPACTION CALCULATOR

Trucking	Roller Model:	[*Click to Select Another Model*]	[**CB534D**]	
Paver Speed	Course Type:	[*Click to Select Course Type*]	[**Binder Course 0/12 (Bc2) - 12 to 15 cm**]	
Compaction	**General Inputs**			
Material Used	Production Rate of Hot Plant:		[**200**]	*tonnes/hr*
Slope	Paving Thickness:		[**125**]	*mm*
	Paving Width:		[**4.00**]	*meter*
Thickness	Actual Drum Width:		[**170**]	*cm*
Job Summary	[*Click to Change Roller Efficiency Rate*]		[**77**]	*%*
Legal	Actual Roller Speed:		[**4**]	*km/h*
	Total Number of Passes:		[**25**]	
	Number of Roller Needed (based on width):		[**1**]	
	Number of Roller Needed (based on speed):		[**4**]	
Exit	**Number of Roller Needed:**		[**4**]	

It is interesting to note how many compactors will be needed when the course type is changed to a thick binder course (Bc2) being laid to a depth of 125 mm. Compaction speed stays the same according to the specification, but the number of passes increases to 25. The calculation shows that 4 compactors with 170 cm wide drums will be needed to match the production rate. And, it is also important to remember that those must be V2 class compactors.

Method specifications are used in many areas and are based on research and project experience. If method specifications are required, that fact will be included in the project plans or in guidelines published by the controlling public works agency. Always consult the public works officials for clarifications if you have questions about method specifications.

End result specifications give more choices to the crews who are doing the work. The second section of Unit 4 deals with the steps for setting up the compaction process to satisfy an end result specification.

Section II: END RESULT SPECIFICATIONS

[DRUM WIDTH]

When selecting compactors for a project, one general rule states that the drums must be wide enough to cover the width of the mat in no more than three overlapping passes. In some situations, the compactor will need more than three passes. If this is the case, another compactor should be added. This general rule applies to projects such as highways, roads and major street paving where production is usually an important factor. The rule typically does not apply for projects like parking lots or low-production projects. The chart below is a reference that is used to help select the correct drum width based solely on covering the mat in no more than three overlapping passes. Drum widths shown are minimum and maximum typically available for higher-production projects.

Required Number of Overlapping Passes per Drum Width

Paving Width Meters / Feet	Drum Widths 140 cm (55")	150 cm (59")	170 cm (67")	200 cm (79")	213 cm (84")
2.5 / 8	2	2	2		
2.75 / 9	3	3	2		
3.00 / 10	3	3	3	2	2
3.35 / 11	3	3	3	2	2
3.70 / 12	(4)	3	3	2	2
4.00 / 13			3	3	2
4.25 / 14			3	3	3
4.50 / 15				3	3
4.80 / 16				3	3
5.20 / 17				3	3
5.50 / 18					3

Note 1: The wider drum models are not recommended on narrower paving widths (less than 3 m / 10') because mat deformation can occur when turning wide drums on narrow mats.

Note 2: Some double drum asphalt compactors feature drum offset. Drum offset significantly increases the width of coverage by the compactor. For example, the Cat CD54 has a drum width of 170 cm (67"). At maximum offset, the CD54 compaction width is 300 cm (118"). Caterpillar recommends drum offset to be used only on mats less than 50 mm (2") thick if you are trying to meet a density specification during the initial phase of compaction.

Drum width is most important for the first compactor operating behind the paver. In general, the compaction process is divided into three phases: initial, intermediate and finish. Different types of compaction equipment and compaction techniques are used on each phase.

[INITIAL COMPACTION]

Initial compaction is the first step in the compaction process and should produce the majority of the target density in the mat. For example, if the target density for final compaction is 95% of theoretical maximum density, the initial phase should produce at least 91% to 93% of theoretical maximum.

Initial compaction should begin at the highest possible mat temperature that supports the weight of the roller without distorting the mat. Remember, once the mat begins to cool, the asphalt cement in the mix gets stiffer and density is harder to achieve.

For this reason, the initial compaction phase must occur in a zone very close to the paver. The paver and initial phase compactor(s) must have the same production rate.

Note: The paving speeds discussed in this section assume the use of vibratory screeds and not screeds which include both tamping and vibration.



The initial phase roller works close to the paver.



Steel drum, vibratory rollers are commonly used for initial compaction.

Steel drum, vibratory rollers are commonly selected for the initial phase of compaction. Because vibratory rollers combine weight and impact, they normally have the highest production rates. Pneumatic compactors are sometimes used in the initial compaction phase on base or binder layers.

Vibratory rollers have different vibratory characteristics as well as different drum widths. How you set up the vibratory system affects how well the initial phase roller will match the paver's production. Let's look at some examples from the Cat Interactive Production Calculator.

PAVER SPEED CALCULATOR

- Trucking
- **Paver Speed**
- Compaction
- Windrow
- Yield
- Slope
- Thickness
- Job Summary
- Legal

- Exit

General Inputs

Paving Thickness:	[2.00]	*in*	[50.8]	*mm*
Paving Width:	[12.00]	*feet*	[3.658]	*meter*
Material Density Uncompacted:	[130]	*lbs/ft³*	[2082]	*kg/m³*

Paver Speed @ Given Production Rate

Production Rate of Hot Plant:	[200]	*tons/hr*	[181]	*tonnes/hr*
Calculated Paving Speed - 100% Efficiency:	[25.6]	*ft/min*	[7.81]	*m/min*
Calculated Paving Speed - 95% Efficiency:	[26.9]	*ft/min*	[8.20]	*m/min*
Calculated Paving Speed - 90% Efficiency:	[28.2]	*ft/min*	[8.59]	*m/min*
Calculated Paving Speed - 85% Efficiency:	[29.4]	*ft/min*	[8.98]	*m/min*
Calculated Paving Speed - 80% Efficiency:	[30.7]	*ft/min*	[9.37]	*m/min*
Calculated Paving Speed - 75% Efficiency:	[32.0]	*ft/min*	[9.76]	*m/min*

Effective Paving Speed: [25.6] *ft/min* [7.81] *m/min*

In this example, the production rate is 181 tonnes per hour (200 tons per hour). The paving width is 3.66 m (12') and the paving thickness is 50 mm (2"). The weight of the material as it passes under the screed (vibratory energy only) is 2082 kg/m³ (130 lb/ft³). If a material transfer device is used on the project, the actual paving speed may be as low as 7.8 m per minute (25.6 feet per minute). If mix is being transferred directly from haul units to the paver, the actual paving speed is calculated at 9.8 meters per minute (32 feet per minute). At a 75% efficiency rate, the effective speed is 7.8 meters per minute (25.6 feet per minute). Therefore, the production rate of the initial compactor must be high enough to match the effective paving speed.



COMPACTION CALCULATOR

- Trucking
- Paver Speed
- **Compaction**
- Windrow
- Yield
- Slope
- Thickness
- Job Summary
- Legal

- Exit

Roller Model: [*Click to Select Another Model*] [CB54]

General Inputs

Paving Width:	[12.00] *feet*	[3.658] *meter*
Actual Drum Width:	[67] *in*	[170.18] *cm*
Amount of Overlap:	[6.0] *in*	[15.2] *cm*
Speed of Vibrator:	[2520] *VPM*	[2520] *VPM*
Impacts (recommended):	[11] *per ft*	[36] *per m*
Number of Passes to Cover Mat Width Once:		[3]
Number of Repeated Passes (from test strips):		[2]
Total Number of Passes:		[7]
Roller Efficiency Rate (recommended 75 to 85%):		[80] *%*

			Effective Paver Speed
Actual Roller Speed:	[229] *fpm*	[70] *mpm*	[25.6] *ft/min*
Effective Roller Speed*:	[26] *fpm*	[8] *mpm*	[7.81] *m/min*

** Effective Roller Speed should be at least 100% but no more than 115% of the Effective Paver Speed.* [**% = 102**]

In this example, a Cat CB54 Asphalt Compactor is available. The CB54 has 170 cm (67") wide drums that can cover the mat in three overlapping passes. Low frequency, 42 Hz (2520 vibrations per minute), has been selected. From work with similar mix on test strips, it is estimated that two repeat passes will produce the target density for the initial phase. Three overlapping passes with two repeat passes creates a seven-pass pattern. An 80% efficiency rate accounts for water refill stops and compactor stops for reversing. An actual working speed of 70 meters per minute (229 feet per minute) will match the effective paving speed. The production calculator also shows that the impact spacing is well within the desired range, which is 26 to 46 impacts per meter (8-14 impacts per foot).



PAVER SPEED CALCULATOR

- Trucking
- **Paver Speed**
- Compaction
- Windrow
- Yield
- Slope
- Thickness
- Job Summary
- Legal

Exit

General Inputs

Paving Thickness:	[2.00]	*in*	[50.8]	*mm*
Paving Width:	[12.00]	*feet*	[3.658]	*meter*
Material Density Uncompacted:	[130]	*lbs/ft³*	[2082]	*kg/m³*

Paver Speed @ Given Production Rate

Production Rate of Hot Plant:	[276]	*tons/hr*	[250]	*tonnes/hr*
Calculated Paving Speed - 100% Efficiency:	[35.4]	*ft/min*	[10.80]	*m/min*
Calculated Paving Speed - 95% Efficiency:	[37.2]	*ft/min*	[11.34]	*m/min*
Calculated Paving Speed - 90% Efficiency:	[38.9]	*ft/min*	[11.88]	*m/min*
Calculated Paving Speed - 85% Efficiency:	[40.7]	*ft/min*	[12.42]	*m/min*
Calculated Paving Speed - 80% Efficiency:	[42.5]	*ft/min*	[12.96]	*m/min*
Calculated Paving Speed - 75% Efficiency:	[44.2]	*ft/min*	[13.50]	*m/min*

Effective Paving Speed: [35.4] *ft/min* [10.80] *m/min*

What if the production rate were higher, say 250 tonnes per hour (276 tons per hour)? The paver speed calculation shows that paving speed increases as hourly tonnage increases, assuming that paving thickness and width do not change. After increasing hourly production to 250 tonnes per hour (276 tons per hour), the effective speed increases to 10.8 meters per minute (35.4 feet per minute).



COMPACTION CALCULATOR

- Trucking
- Paver Speed
- **Compaction**
- Windrow
- Yield
- Slope
- Thickness
- Job Summary
- Legal

- Exit

Roller Model: [*Click to Select Another Model*] [**CB54**]

General Inputs

Paving Width:	[12.00] *feet*	[3.658] *meter*
Actual Drum Width:	[67] *in*	[170.18] *cm*
Amount of Overlap:	[6.0] *in*	[16.5] *cm*
Speed of Vibrator:	[2520] *VPM*	[2520] *VPM*
Impacts (recommended):	[8] *per ft*	[26] *per m*
Number of Passes to Cover Mat Width Once:		[3]
Number of Repeated Passes (from test strips):		[2]
Total Number of Passes:		[7]
Roller Efficiency Rate (recommended 75 to 85%):		[80] *%*

			Effective Paver Speed
Actual Roller Speed:	[315] *fpm*	[96] *mpm*	[35.4] *ft/min*
Effective Roller Speed*:	[36] *fpm*	[11] *mpm*	[10.80] *m/min*

** Effective Roller Speed should be at least 100% but no more than 115% of the Effective Paver Speed.* [**% = 102**]

Next, it is important to determine if the initial phase compactor can keep up with the higher production. The actual working speed of the compactor has to be increased to 96 meters per minute (315 feet per minute) to match the effective paving speed. Keeping the vibratory frequency at 42 Hz (2520 vibrations per minute) while increasing the working speed to 96 meters per minute (315 feet per minute) creates drum impact spacing of 26 impacts per meter (8 impacts per foot). That impact spacing is the minimum allowable. If possible, it would be better to utilize closer impact spacing. One way to affect impact spacing is to increase frequency while keeping working speed the same.



COMPACTION CALCULATOR

- Trucking
- Paver Speed
- **Compaction**
- Windrow
- Yield
- Slope
- Thickness
- Job Summary
- Legal

- Exit

Roller Model: *Click to Select Another Model* [CB54]

General Inputs

Paving Width:	[12.00]	*feet*	[3.658]	*meter*
Actual Drum Width:	[67]	*in*	[170.18]	*cm*
Amount of Overlap:	[6.5]	*in*	[16.5]	*cm*
Speed of Vibrator:	[3800]	*VPM*	[3800]	*VPM*
Impacts (recommended):	[12]	*per ft*	[39]	*per m*
Number of Passes to Cover Mat Width Once:			[3]	
Number of Repeated Passes (from test strips):			[2]	
Total Number of Passes:			[7]	
Roller Efficiency Rate (recommended 75 to 85%):			[80]	*%*

			Effective Paver Speed	
Actual Roller Speed:	[317] *fpm*	[97] *mpm*	[35.4]	*ft/min*
Effective Roller Speed*:	[36] *fpm*	[11] *mpm*	[10.80]	*m/min*

** Effective Roller Speed should be at least 100% but no more than 115% of the Effective Paver Speed.* [% = 102]

The CB54 is often equipped with dual frequency. High frequency on the CB54 is 63.3 Hz (3800 vibrations per minute). With high frequency selected and the working speed set at 97 meters per minute (317 feet per minute), the impact spacing is 39 impacts per meter (12 impacts per foot). That impact spacing is more conducive to producing uniform density and smoothness. Another solution for this example would be to increase drum width so the compactor can cover the mat in two overlapping passes instead of three passes.



COMPACTION CALCULATOR

- Trucking
- Paver Speed
- **Compaction**
- Windrow
- Yield
- Slope
- Thickness
- Job Summary
- Legal

- Exit

Roller Model: [*Click to Select Another Model*] [**CB54 XW**]

General Inputs

Paving Width:	[12.00] *feet*	[3.658] *meter*
Actual Drum Width:	[79] *in*	[200.66] *cm*
Amount of Overlap:	[6.5] *in*	[16.5] *cm*
Speed of Vibrator:	[2520] *VPM*	[2520] *VPM*
Impacts (recommended):	[11] *per ft*	[36] *per m*
Number of Passes to Cover Mat Width Once:		[2]
Number of Repeated Passes (from test strips):		[2]
Total Number of Passes:		[5]
Roller Efficiency Rate (recommended 75 to 85%):		[80] *%*

			Effective Paver Speed
Actual Roller Speed:	[229] *fpm*	[70] *mpm*	[35.4] *ft/min*
Effective Roller Speed*:	[37] *fpm*	[11] *mpm*	[10.80] *m/min*

** Effective Roller Speed should be at least 100% but no more than 115% of the Effective Paver Speed.* [**% = 105**]

A compactor with 200 cm (79") drum width, if available, will cover the 3.66 m (12') mat in two overlapping passes instead of three passes. The extra drum width changes the rolling pattern from seven passes to five passes. Working speed deceases to 70 meters per minute (229 feet per minute). The vibratory frequency reverts back to 42 Hz (2520 vibrations per minute). That working speed produces more acceptable impact spacing, 36 impacts per meter (11 impacts per foot).

Remember, low frequency is always associated with higher amplitude. You will generally reach target density faster using low frequency and medium to high amplitude on mats that are 50 mm (2") and thicker.

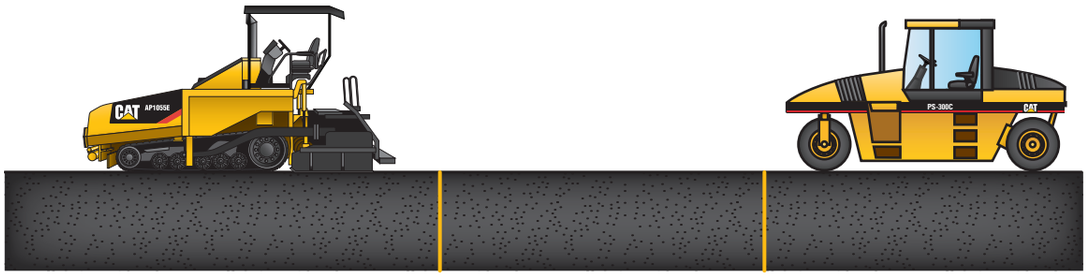
Selecting the correct compactor for the initial phase of compaction is critical to achieving consistent and acceptable density. Pre-project planning is needed to confirm that the compactor will be able to match the paving speed.

[INTERMEDIATE COMPACTION]

Intermediate compaction follows immediately after the initial phase on most mixes. The goals of intermediate compaction are to create the final target density in the mat and to begin the process of cleaning up marks left by the initial phase compactor.

The mat should be hot enough to allow some aggregate movement so usually intermediate compactors work in the temperature zone just behind the initial phase temperature zone. Depending on the thickness of the mat, you have to be careful about the type and amount of force you apply. Remember, the mat in the final phase is already close to the final target density and you're trying to increase density by 1% to 3%.

Pneumatic compactors are a common choice for intermediate compaction because they can exert high static pressure without delivering impact forces. The type of pneumatic compactor selected for the intermediate phase depends on mat thickness and asphalt mix formula.

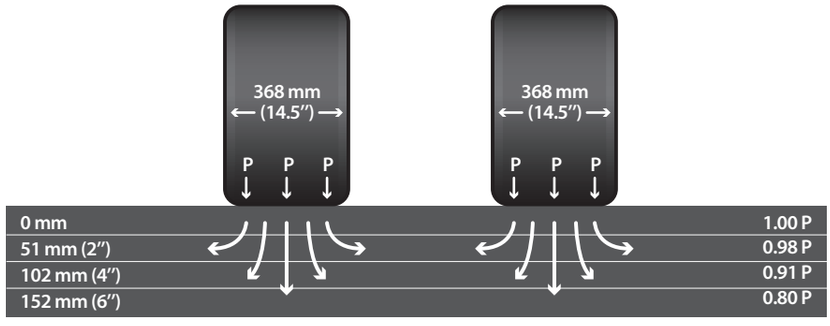


Intermediate compaction closely follows the initial phase.



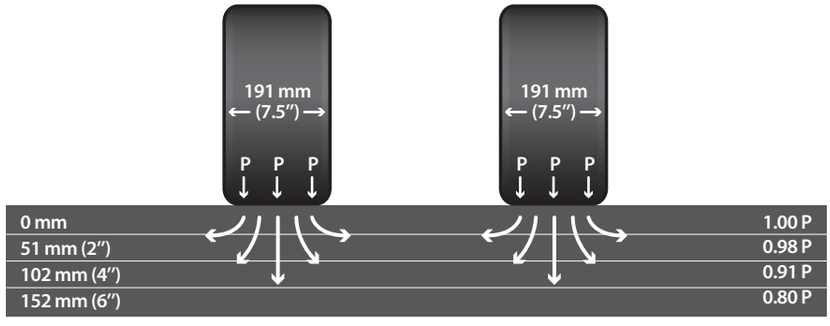
Pneumatic compactors can exert high forces without delivering impact forces.

WIDE-TIRE PRESSURE DISTRIBUTION



For thick, harsh mat compaction, a pneumatic compactor with wide tires is the best choice. The wide tires can withstand much higher loads that are needed to get final density in harsh mixes that are characterized by large aggregates. Notice that even at a depth of 100 mm (4") the compaction pressure is still 90% effective. Wide-tire pneumatic compactors are a good choice for base and binder layers that are normally the thickest layers in the pavement structure.

NARROW-TIRE PRESSURE DISTRIBUTION

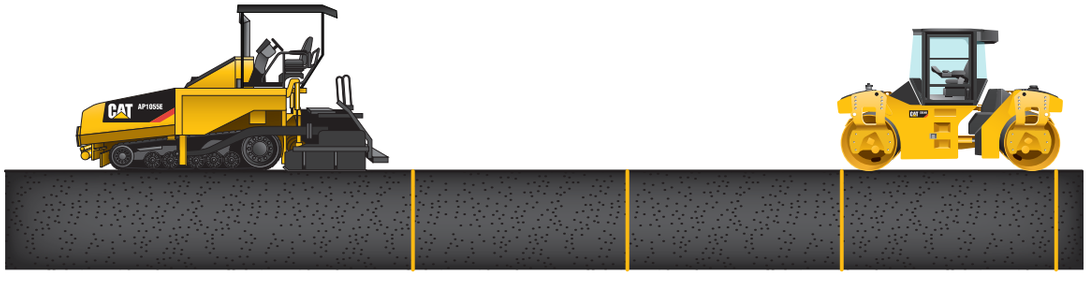


Pneumatic compactors with narrow tires exert high ground contact pressure, but that pressure is most effective on thin, stiff mats. Notice that the pressure diminishes rapidly when mat depth exceeds 50 mm (2"). Pneumatic compactors with narrow tires are a good choice for wear layers that are typically the thinnest and made of the stiffest material.

If steel drum vibratory rollers are selected for the intermediate phase of compaction, be careful not to use too much impact force. The mat is already close to final density so high frequency combined with the lowest amplitude is probably going to work the best. Also, the mat temperature in the intermediate zone will have cooled to around 110° C (230° F). Vibrating aggressively on cool mats can result in impact marks that do not clean up during the finish phase or can cause fractured aggregates at the surface of the mat.



Vibratory rollers used in the intermediate phase are set to low amplitude.



The finish roller may be an hour behind the paver where the mat is cool.

[FINISH COMPACTION]

The final phase is finish compaction. The primary goal of the finish phase is to remove any drum stop marks or pneumatic tire marks. There may be small gains in density during the finish phase, but if you depend on getting more density during the finish phase, you are taking a risk.

The finish phase normally takes place when the mat is still warm enough to allow the removal of marks in the surface. If the finish compactor is leaving its own stop marks, the mat is still too warm and finish compaction should be delayed even more. It is common for the finish compactor to be as much as an hour behind the paver. Finish compactor operators benefit from on-board temperature sensors or hand-held temperature scanners to help them stay in the correct temperature zone.